Vectors and Vector-Valued Functions

Section 1: 3D Vectors, Dot and Cross Products

1 Given $\mathbf{u} = \langle 2, -1, 3 \rangle$ and $\mathbf{v} = \langle 4, 2, -1 \rangle$, find $\mathbf{u} \cdot \mathbf{v}$.

a. 3

b. 7

c. 13

d. -1

2 Find the cross product $\mathbf{u} \times \mathbf{v}$ for $\mathbf{u} = \langle 1, 0, 2 \rangle$ and $\mathbf{v} = \langle 0, 3, -1 \rangle$.

a. $\langle -6, 1, 3 \rangle$

b. $\langle 6, -1, -3 \rangle$

c. $\langle -6, -1, 3 \rangle$

d. $\langle 6, 1, 3 \rangle$

3 Find the area of the parallelogram determined by $\mathbf{u} = \langle 1, 2, 0 \rangle$ and $\mathbf{v} = \langle 0, 3, 4 \rangle$.

a. $\sqrt{89}$

b. 9

c. $\sqrt{73}$

d. 10

Section 2: Derivatives and Integrals of Vector-Valued Functions

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4 Given $\mathbf{r}(t) = \langle t^2, \sin(2t), e^{3t} \rangle$, find $\mathbf{r}'(t)$.

a. $\langle 2t, 2\cos(2t), 3e^{3t} \rangle$

b. $\langle 2t, \cos(2t), e^{3t} \rangle$

c. $\langle t, 2\cos(2t), 3e^{3t} \rangle$

d. $\langle 2t, \ -2\cos(2t), 3e^{3t} \rangle$

5 Find $\int_0^1 \langle 2t, 3t^2, 4t^3 \rangle \, dt$.

a. $\langle 1, 1, 1 \rangle$

b. $\langle 2, 3, 4 \rangle$

c. $\langle 1/2, 1/3, 1/4 \rangle$

d. $\langle 1, 2, 3 \rangle$

6 Find the unit tangent vector $\mathbf{T}(t)$ to $\mathbf{r}(t) = \langle 3\cos t, 3\sin t, 4t \rangle$.

a. $\langle -\frac{3}{5}\sin t, \frac{3}{5}\cos t, \frac{4}{5} \rangle$

b. $\langle -3\sin t, 3\cos t, 4 \rangle$

c. $\langle -\frac{3}{4}\sin t, \frac{3}{4}\cos t, 1 \rangle$

d. $\frac{1}{5}\langle 3\cos t, 3\sin t, 4t \rangle$

Section 3: Arc Length and Curvature of Space Curves

7 Find the arc length of the circular helix $\mathbf{r}(t) = \langle 3\cos t, 3\sin t, 4t \rangle$ for $0 \le t \le 2\pi$.

a. 10π

b. 5π

c. 20π

d. 6π

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8 State the formula for curvature $\kappa(t)$ of a space curve in terms of $\mathbf{r}'$ and $\mathbf{r}''$.

a. $\kappa(t) = \dfrac{\|\mathbf{r}'(t) \times \mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|^3}$

b. $\kappa(t) = \dfrac{\|\mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|^2}$

c. $\kappa(t) = \dfrac{\mathbf{r}'(t) \cdot \mathbf{r}''(t)}{\|\mathbf{r}'(t)\|^3}$

d. $\kappa(t) = \|\mathbf{r}''(t)\|$

9 Find the curvature of a circle of radius R.

a. $\kappa = 1/R$

b. $\kappa = R$

c. $\kappa = 2\pi R$

d. $\kappa = 1/R^2$

Section 4: Velocity, Acceleration, and Projectile Motion

10 If position is $\mathbf{r}(t) = \langle t^2, 4t, -16t^2 + 64t \rangle$, find the velocity and acceleration vectors at $t = 1$.

a. $\mathbf{v}(1) = \langle 2, 4, 32 \rangle$, $\mathbf{a}(1) = \langle 2, 0, -32 \rangle$

b. $\mathbf{v}(1) = \langle 1, 4, 48 \rangle$, $\mathbf{a}(1) = \langle 2, 0, -16 \rangle$

c. $\mathbf{v}(1) = \langle 2, 4, 0 \rangle$, $\mathbf{a}(1) = \langle 2, 0, -32 \rangle$

d. $\mathbf{v}(1) = \langle 2, 4, -32 \rangle$

11 Find the speed at $t = 0$ for a particle with velocity $\mathbf{v}(t) = \langle 3\cos t, 4, 3\sin t \rangle$.

a. 5

b. 7

c. 4

d. $\sqrt{7}$

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12 Decompose acceleration into tangential and normal components: $\mathbf{a} = a_T\mathbf{T} + a_N\mathbf{N}$. What is a_T?

a. $a_T = \frac{\mathbf{v} \cdot \mathbf{a}}{\|\mathbf{v}\|}$

b. $a_T = \frac{\|\mathbf{v} \times \mathbf{a}\|}{\|\mathbf{v}\|}$

c. $a_T = \|\mathbf{a}\|$

d. $a_T = \frac{\mathbf{a}}{\|\mathbf{v}\|}$

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